Experiment - 1

History of internet

1. web client and web browser
The web client is the general term for any software or device that requests services from a server. A web browser is the most prominent type of web client

function :- The browser acts as the user interface, interpreting raw code and assets (HTML, CSS, javascript, image) received from a server and assembling them into the visuals, interactive webpages you see

components of a web browser :-
- user interface
- Rendering Engine
- Browser Engine
- Networking

2. DNS (Domain Name System)
DNS is the internet's distributed service, translating human-readable domain names into machine-readable IP address.

- function :- without DNS, users would have to remember complex numerical IP address for every website.

- The Resolution process :-
- user input
- chache check
- Resolver query
- Hierarchical query
- IP Address Returned

3. web server
The webserver is a power full computer hardware

and software) that stores website files and delivers them to clients.

- function:- it listens for HTTP requests process them and send back the requested resource via HTTP/HTTPS. it uses protocols like HTTP, FTP, and SMTP to manage data transfer.

- request handling
- static Vs Dynamic content.

4. Search Engines:
Search engines are tools that helps users discover information across the vast expanse of the web. They are not directly involved in the real time client-server

- function:- They organize web content to efficiency match user queries with relevent web pages.

- core processes
- Crawling.
- indexing.
- Ranking.

flow

initialization by web client

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Translating The Name via DNS

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The Request to webserver

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processing to the webserver

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Rendering by web client

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Result